

DATA266 Laboratory 1: Collaborative Report on Language Modeling, Sentiment Classification, and Unpaired Image-to-Image Translation

San José State University — Fall 2026

Course	DATA266
Assignment	Laboratory 1 (Team Report)
Team number	Lab Pair 17
Repository	https://github.com/pramodsatyad/data266-lab1-pair17

Authors

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Abstract

This report presents the final Team 17 results for DATA 266 Laboratory 1. Each member independently implemented and evaluated models for three tasks: a character-level GPT trained from scratch on TinyStories, binary sentiment classification on Yelp Polarity using embeddings learned from scratch, and unpaired Monet-photo translation with CycleGAN. The report compares architecture and hyperparameter choices, reports the required quantitative results, summarizes observed failure modes, and links each claim to code, checkpoints, metrics, logs, and generated outputs in the shared repository. The confirmed CycleGAN Kaggle public scores are -50.257851 for Pramod Dindukurthi and -49.0329 for Manav Patel.

1. Introduction

Laboratory 1 follows a practical research workflow: individual experimentation first, followed by a team-level comparison of model quality, efficiency, failure behavior, and reproducibility. The three tasks cover autoregressive language modeling, supervised text classification, and unpaired image-to-image translation. All implementations were trained independently so that the comparison reflects different design choices rather than duplicated models.

Task 1 tests causal self-attention and autoregressive decoding without prebuilt Transformer modules. Task 2 compares six classifiers under the constraint that all textual embeddings are learned from scratch. Task 3 compares two independently trained CycleGAN systems on the Monet-photo domains and records the final class-competition submissions. The sections below separate methodology, quantitative results, error analysis, and joint interpretation, with repository paths provided for traceability.

2. Team Ownership and Reproducibility

Both members independently implemented, trained, and evaluated their own models for all three tasks. **Pramod Dindukurthi** built a compact 4-layer character-level GPT, a TextCNN/DPCNN/scratch-Transformer sentiment suite, and a CycleGAN trained for 350,704 main updates followed by controlled identity-loss pilots. **Manav Patel** built a larger 6-layer character-level GPT, a mean-embedding/BiLSTM/BiLSTM-attention sentiment suite, and a CycleGAN trained through 100,000 steps with DiffAugment and EMA checkpointing. The team report below compares these independent choices, preserves protocol differences where they exist, and points back to the repository artifacts used as evidence.

The final repository contains both members' folders under all three task directories. Environment manifests and retained raw logs are stored under reproducibility/, while each member directory contains source code, processed metadata, checkpoints, outputs, metrics_report.csv, failure_analysis.md, and results.md. The repository root README documents the smoke-test and reproduction entry points, and the combined report is stored under report/.

3. Task 1 - Character-Level Language Modeling on TinyStories

3.1 Methods and Training Setup

Both models used character-level autoregressive language modeling on TinyStories and implemented causal self-attention manually, without prebuilt Transformer or attention modules. The key difference was scale: Pramod used a smaller 4-block, 256-dimensional model with a 128-character context, while Manav used 6 blocks, 384 dimensions, and a 256-character context.

3.2 Quantitative Results

Member	Architecture	Key hyperparameters	Params	Train CE	Val CE	PPL	BPC	Gen. gap	Top-1	D-1 / D-2 / D-3	Rep. 4-gram	Stability	Train tok/s	Gen tok/s	Peak MB	Train time
Pramod	4 blocks, 8 heads, d=256, FFN=1024, ctx=128	bs=128, 10 ep, AdamW, lr=4e-4, dropout=.1, warmup+cosine	3.22M	0.6758	0.6852	1.984	0.989	0.0094	0.7813	0.068 / 0.320 / 0.532	0.3596	20 non-finite grad-norm events; no loss NaNs	988,113	424.1	1,335	16.0 min
Manav	6 blocks, 6 heads, d=384, FFN=1536, ctx=256	bs=64, 10 ep, AdamW, lr=6e-4, wd=.1, bf16, clip=1.0	10.83M	0.5388	0.5332	1.704	0.769	-0.0056	0.8289	0.294 / 0.676 / 0.823	0.1270	0 NaNs, 0 grad spikes; max grad 10.04 before clipping	344,360	95.75	3,925	45.3 min

Evidence: [Pramod results](#) | [Pramod metrics](#) | [Manav results](#) | [Manav manifest](#)

3.4 Discussion

Manav's larger context and wider/deeper network produced the stronger language-modeling metrics: validation cross-entropy fell from 0.6852 to 0.5332 and top-1 next-character accuracy rose from 78.1% to 82.9%. The generated text was also less repetitive by the repeated 4-gram metric. The cost of that improvement was clear: the model had about 3.4x as many parameters, used roughly 3x the peak GPU memory, and took about 2.8x as long to train. Pramod's compact model processed substantially more training tokens per second and generated characters faster, making it the more efficient reference implementation. Both models still showed the same qualitative weakness: local syntax was convincing, but long-range story consistency was fragile.

3.3 Failure Analysis

Pramod - representative failure cases

Type	Prompt	Actual generated snippet	Interpretation
Repetition	Once upon a time	"She wanted to make a colorful colorful colors ... She wanted to make a colorful colorful colors ..."	Greedy decoding fell into a locally probable phrase loop.
Entity drift	There was a small dragon	"There was a small dragon named Daisy ... Lily was so happy ... Tom said."	Sampling improved diversity but weakened character consistency.
Contradiction	Tom went to the forest	"It is a bad dog. It is not a bad dog. It is a bad dog."	The model maintained local grammar but contradicted itself globally.

Evidence: [Pramod full failure analysis](#)

Manav - representative failure cases

Type	Prompt	Actual generated snippet	Interpretation
Repetition	It was a sunny day	"The birds were singing and the sun was shining." repeated multiple times.	Greedy decoding had no randomness to escape the loop.
Entity/object confusion	Once upon a time	"There was a thick car named Daisy ... she ran to the car ... 'Hi, Daisy!'"	The character identity changed within the short context window.
Prompt abandonment	The dog was	The model ended the dog story, emitted < endoftext >, and started a new Lily story.	The end-of-story marker strongly triggered a new training-pattern continuation.

Evidence: [Manav full failure analysis](#)

4. Task 2 - Yelp Polarity Sentiment Classification

4.1 Methods and Model Configurations

Each member trained three models with embeddings learned from scratch. Pramod compared convolutional and Transformer-style encoders; Manav compared a bag-of-embeddings baseline with recurrent models. This creates a useful team-level comparison between local n-gram modeling, sequential recurrence, attention pooling, and a scratch Transformer.

Member / model	Architecture	Key training settings
Pramod - TextCNN	192-d embeddings; Conv1D kernels 3/4/5; 160 filters each; max-pool classifier	max_len=256; dropout=0.4; lr=1.5e-3; up to 6 epochs; AdamW + OneCycleLR; grad clip=1.0
Pramod - DPCNN	192-d embeddings; 192 channels; three residual Conv1D blocks with downsampling	max_len=256; dropout=0.3; lr=1.0e-3; up to 6 epochs; AdamW + OneCycleLR; grad clip=1.0
Pramod - Transformer	256-d token/position embeddings; 3 encoder layers; 8 heads; FFN=768	max_len=256; dropout=0.15; lr=3e-4; up to 8 epochs; AdamW + OneCycleLR; grad clip=1.0
Manav - Mean embedding	128-d learned word embeddings; masked mean pooling; linear classifier	vocab=30k; max_len=256; dropout=0.2; lr=2e-3; batch=256; 5 epochs; wd=0.01
Manav - BiLSTM	128-d embeddings; bidirectional LSTM, hidden=128, 1 layer	max_len=256; dropout=0.3; lr=1e-3; batch=256; 5 epochs; wd=0.01; grad clip=1.0
Manav - BiLSTM + attention	128-d embeddings; 2-layer bidirectional LSTM, hidden=128; attention pooling	max_len=256; dropout=0.3; lr=1e-3; batch=256; 5 epochs; wd=0.01; grad clip=1.0

4.2 Quantitative Results

Member	Model	Accuracy %	Macro-F1 %	ROC-AUC %	PR-AUC %	MCC %	Brier	ECE	Params	Train s	Examples/s	Peak MB
Pramod	TextCNN	93.44	93.44	98.28	98.32	86.89	0.0494	0.0202	9.97M	46.7	24,410	393
Pramod	DPCNN	92.97	92.97	98.20	98.26	85.95	0.0513	0.0094	10.38M	37.2	20,427	538
Pramod	Scratch Transformer	92.01	92.01	97.63	97.73	84.03	0.0590	0.0069	14.84M	158.0	6,013	3,268
Manav	Mean-embedding baseline	92.87	92.87	97.71	97.70	85.74	0.0540	0.0087	3.84M	9.5	103,107	140
Manav	BiLSTM	94.26	94.26	98.62	98.67	88.53	0.0437	0.0184	4.10M	147.5	6,570	364

Member	Model	Accuracy %	Macro-F1 %	ROC-AUC %	PR-AUC %	MCC %	Brier	ECE	Params	Train s	Examples/s	Peak MB
Manav	2-layer BiLSTM + attention	94.20	94.20	98.66	98.70	88.41	0.0439	0.0184	4.50M	380.2	2,533	520

Model	Macro P	Macro R	Micro P/R/F1	Weighted P	Weighted R	Weighted F1	95% Acc CI	95% Macro-F1 CI	95% MCC CI	McNemar comparison
Pramod TextCNN	0.9344	0.9344	0.9344	0.9344	0.9344	0.9344	[0.9320, 0.9367]	[0.9318, 0.9369]	[0.8637, 0.8742]	vs DPCNN p=1.25e-4; vs Transformer p≈0
Pramod DPCNN	0.9298	0.9297	0.9297	0.9298	0.9297	0.9297	[0.9271, 0.9320]	[0.9268, 0.9323]	[0.8542, 0.8645]	Compared against TextCNN
Pramod Transformer	0.9202	0.9201	0.9201	0.9202	0.9201	0.9201	[0.9174, 0.9227]	[0.9173, 0.9229]	[0.8351, 0.8458]	Compared against TextCNN
Manav baseline	0.9287	0.9287	0.9287	0.9287	0.9287	0.9287	[0.9262, 0.9313]	[0.9262, 0.9313]	[0.8525, 0.8626]	Reference baseline
Manav BiLSTM	0.9426	0.9426	0.9426	0.9426	0.9426	0.9426	[0.9401, 0.9450]	[0.9401, 0.9450]	[0.8803, 0.8901]	vs baseline p=1.98e-33
Manav BiLSTM+attn	0.9421	0.9420	0.9420	0.9421	0.9420	0.9420	[0.9397, 0.9443]	[0.9397, 0.9442]	[0.8794, 0.8885]	vs baseline p=2.89e-29

Model	Short F1 / err	Medium F1 / err	Long F1 / err	Negation F1 / err	Contrast F1 / err
Pramod TextCNN	0.9266 / 7.02%	0.9351 / 6.49%	0.9343 / 6.18%	0.9316 / 6.61%	0.9276 / 7.18%
Pramod DPCNN	0.9252 / 7.14%	0.9316 / 6.84%	0.9211 / 7.43%	0.9255 / 7.22%	0.9194 / 8.00%
Pramod Transformer	0.9164 / 8.03%	0.9212 / 7.88%	0.9115 / 8.25%	0.9123 / 8.44%	0.9081 / 9.09%
Manav baseline	0.9248 / 7.21%	0.9277 / 7.23%	0.9278 / 6.77%	0.9208 / 7.64%	0.9189 / 8.01%
Manav BiLSTM	0.9385 / 5.90%	0.9440 / 5.60%	0.9369 / 5.92%	0.9396 / 5.82%	0.9351 / 6.41%
Manav BiLSTM+attn	0.9355 / 6.19%	0.9423 / 5.77%	0.9421 / 5.42%	0.9398 / 5.80%	0.9357 / 6.35%

Evidence: [Pramod results](#) | [Pramod metrics](#) | [Manav results](#) | [Manav metrics](#)

4.3 Discussion

The strongest single model was Manav's BiLSTM at 94.26% accuracy and macro-F1, narrowly ahead of his attention model and Pramod's TextCNN. The attention layer did not deliver a reliable overall gain over the plain BiLSTM despite more than doubling training time, although it was strongest on long reviews. Pramod's TextCNN was the best of his three models and was much more efficient than his scratch Transformer; the Transformer used the most memory and training time while giving the lowest accuracy. Across both members, contrast-heavy and negation-heavy reviews were consistently among the hardest slices. This agrees with the manual error reviews: late sentiment reversals, mixed aspects, sarcasm, and temporal context were common failure modes. A practical next experiment would be sentence- or clause-level aggregation with explicit treatment of contrast markers such as “but” and “however”, while monitoring calibration so that accuracy gains do not simply make confidence more extreme.

4.4 Error Analysis

The repository contains the full 20-example review for each member. The table below keeps representative cases in the combined report so the main failure patterns remain visible without duplicating several pages of raw text. The full 20-case analyses are linked immediately after the table.

Member	Category	Actual review excerpt	Observed failure	Testable direction
Pramod	Confident FP	"I love cafe rio!... However ... DO NOT EAT AT THIS LOCATION!"	Late contrast reversal; positive opening dominates.	Contrast-aware or position-aware pooling.
Pramod	Confident FP	"Yay. Cafeteria food. Yay."	Sarcasm makes positive cue words misleading.	Sarcasm/incongruity augmentation.
Pramod	Confident FN	"This place is so much better since they changed owners... it was terrible... Now its much better."	Temporal reversal; long old negative history overwhelms current sentiment.	Hierarchical sentence/discourse modeling.
Pramod	Near threshold	"Oh my god the food is extremely good... even though ... they charge you the soft drink..."	Mixed aspects place the example near the decision boundary.	Clause-level aggregation and threshold calibration.
Manav	Confident FP	"Wow love the place ... Great place to come and relax..." (label 0)	Likely label noise; wording is clearly positive.	Audit label-text disagreements / robust loss.
Manav	Confident FN	"This place is so much better since they changed owners..."	Before/after structure; visible text contains long negative history.	Discourse-aware modeling.
Manav	Near threshold	"AVOID this company. Very slow service... 0 stars..."	Stacked negation/rare emphatic patterns still landed near 0.5.	Targeted negation examples / calibration.
Manav	Slice error	"A small coffee is now \$2.50 Lol."	Very short review with little explicit sentiment signal.	Short-review handling and richer context features.

Evidence: [Pramod full 20-case review](#) | [Manav full 20-case review](#)

5. Task 3 - Unpaired Image-to-Image Translation (CycleGAN)

5.1 Methods and Training Configuration

Both members implemented the same broad CycleGAN family (two ResNet generators and two PatchGAN discriminators), but the training strategy differed. Pramod trained a long 350,704-update main run and then ran controlled identity-loss pilots. Manav trained in three stages to 100,000 steps and used DiffAugment and EMA checkpointing. The evaluation protocols in the current repository are not fully aligned, so the numbers below are reported faithfully but should not be interpreted as a perfectly controlled cross-member benchmark.

Item	Pramod Dindukurthi	Manav Patel
Generators / discriminators	Two 9-block, 64-channel ResNet generators; two 70x70 PatchGAN discriminators	Two 9-block ResNet generators; two 70x70 PatchGAN discriminators
Core losses	LSGAN adversarial; cycle L1 weight=10; identity L1	LSGAN adversarial; cycle L1 weight=10; identity L1 weight=5
Main optimization	batch=1; Adam lr=2e-4, betas=(0.5,0.999); replay pool=50; EMA=0.999; seed=42	batch=1; Adam lr=2e-4; replay pool=50; EMA=0.999; DiffAugment
Training schedule	350,704 main updates; then two 20k continuation arms; selected id=2.5 at +5k with lr=5e-5	three stages ending at 100,000 steps; linear LR decay with staged continuation
Selection	controlled identity-loss pilot; selected raw checkpoint by official-style evaluation	selected step 100,000 EMA checkpoint after staged metric improvement
Hardware	NVIDIA RTX 4090	NVIDIA RTX 4090

5.2 Quantitative Results

Member	Selected checkpoint	A2B FID	B2A FID	KID A2B / B2A	Density or precision	Coverage or recall	Cycle L1 A/B	LPIPS A/B	Content cosine A/B	Kaggle public
Pramod	raw id=2.5, +5k after 350,704 main updates	161.246 held-out	165.132 held-out	0.02316 / 0.01532	density 0.604 / 0.757	coverage 0.104 / 1.000	0.0672 / 0.0458	0.2744 / 0.4364	0.8376 / 0.7127	-50.257851
Manav	EMA step 100,000	88.442 full-set	88.195 full-set	0.0243 / 0.0141	precision 0.573 / 0.387	recall 0.300 / 0.600	0.0741 / 0.0916	0.2686 / 0.3690	0.8374 / 0.7799	-49.0329

Training and efficiency summary

Member	Architecture / training	Stability	Parameters	Training time	Throughput	Peak memory
Pramod	9-block 64ch ResNet generators; 70x70 PatchGAN; LSGAN; cycle=10; identity pilot 5→2.5	selected pilot NaN=0, Inf=0	28.29M	10.69 h update time	17.46 images/s (selected pilot)	2.52 GiB allocated
Manav	9-block ResNet generators; 70x70 PatchGAN; DiffAugment; replay pool=50; EMA=.999	0 NaNs; controlled losses and gradients	28.29M	2.41 h	23.35 images/s	10.95 GB

Protocol note. Pramod's table above uses his held-out local diagnostic protocol for FID/KID/LPIPS/content metrics; his separate official-style evaluator reported mean FID 100.104527, MiFID 0.411176, and score -50.257851 on 300 images per direction, matching his confirmed Kaggle public score. Manav's reported local FID/MiFID values come from his full-set evaluation. Because these pipelines use different sample definitions and evaluators, the raw local FID values should not be ranked against each other without rerunning both checkpoints through one shared evaluator. The confirmed Kaggle public scores from the final submissions are -50.257851 for Pramod and -49.0329 for Manav.

Evidence: [Pramod results](#) | [Pramod full metrics](#) | [Manav results](#) | [Manav full metrics](#)

5.3 Discussion

The two runs show different stabilization strategies. Pramod used long training plus a controlled identity-loss ablation, selecting the identity=2.5 arm at +5,000 updates. Manav used augmentation, EMA, and staged continuation. Their confirmed Kaggle public scores were -50.257851 and -49.0329, respectively. Both failure analyses show that cycle consistency alone does not prevent color shifts, blur, or texture artifacts, so the next team comparison should rerun both checkpoints with one shared evaluator and the same fixed sample set.

5.4 Failure Modes

Member	Representative failure / limitation	What it shows	Proposed next step
Pramod	Selected model improved both directional FID/KID numerically, but content cosine declined slightly and B2A cycle error increased.	Automatic metrics can move in different directions and do not prove perceptual improvement.	Use the same fixed evaluation sample set and shared evaluator for a cleaner cross-member comparison.
Manav	Smooth sunset and lightning images showed checkerboard/grid artifacts and large color shifts. Fine bird textures blurred despite reasonable content cosine.	Transposed-convolution upsampling and a small Monet domain can hurt smooth regions and high-frequency detail.	Resize-then-convolution upsampling; stronger color consistency; evaluate same worst cases again.

Evidence: [Pramod Task 3 failure analysis](#) | [Manav Task 3 failure analysis](#)

6. Reproducibility and Evidence Trail

The final team repository contains both members' implementations for all three tasks, together with member-specific metrics, failure analyses, checkpoints, outputs, and shared reproducibility artifacts. Manav retains explicit manifests and raw logs for Tasks 1-3. Pramod retains a Task 1 manifest/log, a post-run Task 2 manifest plus the executed notebook, checkpoints and structured experiment outputs, and detailed Task 3 manifests/raw logs with checkpoint hashes and pilot histories. Pramod's Task 2 runner was configured to write a raw master log during execution, but that console stream was not retained during packaging; the completed run remains traceable through the notebook, three trained checkpoints, experiment summary, metric and slice reports, plots, and error-analysis artifacts.

Area	Pramod	Manav	Evidence / note
Task 1	results, metrics, failure analysis, hardware manifest, training log/history	results, metrics, failure analysis, manifest, raw log	Both runs traceable; checkpoints retained.
Task 2	3 checkpoints, metrics, plots, experiment summary, notebook/runner; raw master log not retained during packaging	3 checkpoints, manifest, raw logs, predictions, plots	Retained evidence includes the executed notebook, checkpoints, experiment summary, metrics, plots, and error-analysis outputs.
Task 3	extensive raw logs/manifests, SHA-256 selected checkpoint provenance, run guide, plots	manifest, raw logs, staged checkpoints, evaluation outputs	Pramod large selected checkpoint is not in normal Git; retrieval needs durable external storage.
Final report	DATA266_Lab1_Report_Team_17.pdf	Final team report	Stored under report/ with the root README and merged member folders.

The repository root README documents environment setup, a one-command Task 3 smoke test, task-specific reproduction commands, dataset expectations, and the location of each member's results. The combined report is stored at report/DATA266_Lab1_Report_Team_17.pdf. No credentials, API keys, or hard-coded personal paths are required by the documented reproduction flow.

7. Conclusions

The three tasks reinforced the same engineering lesson: the best model is not automatically the largest one. In Task 1, additional depth and context improved modeling quality, but the compact GPT was substantially cheaper. In Task 2, the best overall classifier was a plain BiLSTM, while the attention variant was nearly tied at much higher cost and the scratch Transformer underperformed the simpler convolutional baselines. In Task 3, both CycleGANs trained stably, but qualitative

failures remained even when automatic metrics improved. The comparison also showed why evaluation discipline matters: protocol differences can change how a score should be interpreted, so the final report keeps those differences explicit rather than forcing a misleading direct ranking. Across all three tasks, the most useful conclusions came from balancing predictive quality, efficiency, failure behavior, and reproducibility rather than relying on a single headline metric.

8. References

Vaswani, A., et al. (2017). Attention Is All You Need. *Advances in Neural Information Processing Systems* 30.

Eldan, R., & Li, Y. (2023). TinyStories: How Small Can Language Models Be and Still Speak Coherent English? *arXiv:2305.07759*.

Zhu, J.-Y., Park, T., Isola, P., & Efros, A. A. (2017). Unpaired Image-to-Image Translation Using Cycle-Consistent Adversarial Networks. *ICCV*.

Appendix A - Artifact Index

The paths below are the primary evidence locations used to trace the report tables back to member-level artifacts in the final repository.

Task	Pramod	Manav
Task 1	task1_llm/Pramod_Dindukurthi/{results.md, metrics_report.csv, failure_analysis.md}	task1_llm/member_manav/{results.md, metrics_report.csv, failure_analysis.md}
Task 2	task2_sentiment/Pramod_Dindukurthi/{results.md, metrics_report.csv, failure_analysis.md, outputs/}	task2_sentiment/member_manav/{results.md, metrics_report.csv, failure_analysis.md, outputs/gpu/}
Task 3	task3_gan/Pramod_Dindukurthi/{results.md, full_metrics_report.csv, failure_analysis.md, evidence/, experiments/}	task3_gan/member_manav/{results.md, full_metrics_report.csv, failure_analysis.md, outputs/gpu/}
Reproducibility	reproducibility/manifests/task3_Pramod_* and reproducibility/raw_logs/task3_Pramod_*	reproducibility/manifests/manav_task*.txt and reproducibility/raw_logs/manav_task*_*.txt